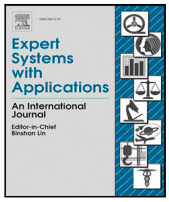
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Predicting diversification scores of videos in recommendation network

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ABSTRACT

Recommendation systems are crucial in various platforms from shopping to online social networks. A key challenge in recommendation systems is to leverage diversity, exposing or recommending diverse items to individuals. Despite much effort on studying diversity in the recommendation systems, little work has focused on estimating how much an item will potentially affect user's diversity experiences by contributing to consecutive recommendations in a session. In this work, we propose a deep learning model that can predict *diversification scores*, which is a degree of potential contribution to users' diversity experiences of an item. The proposed model adopts multiple graph neural network layers with a novel attention mechanism that can capture the features of a given item and its related items in terms of recommendation. To prove the effectiveness of our approach, we collect a large dataset of video recommendations from YouTube and conduct random-walk experiments to simulate user traces. The evaluation results on the dataset shows that the proposed model accurately predicts each item's contribution on user diversity experiences.

1. Introduction

Recommendation systems have become a key pillar in various online service platforms from online shopping sites to online social networks. In such platforms, a recommendation system considers the information related to target users including interests, geographical characteristics, or personal traits to suggest items that a user might be interested in. Relieving users of the overhead of searching from a wide array of items (Hanani et al., 2001), recommendation systems have been widely adopted in popular online services such as Amazon (Smith & Linden, 2017), Netflix (Gomez-Uribe & Hunt, 2016), and YouTube (Beutel et al., 2018).

Recently, not only relevancy of recommended items to users' interests, but *diversity*, indicating how diverse items are recommended to or experienced by users, has been considered as an important feature of recommendation systems. In particular, prior studies have warned that recommendation systems that do not consider diversity may restrict the information exposure and cause well-known social problems such as filter bubbles (Pariser, 2011) or echo chambers (Nguyen et al., 2014).

With the growing importance of diversity in recommendation systems, there has been much effort to enhance diversity from both service provider and user perspectives. From a provider perspective, suggesting underprovisioned items has mainly been focused on, while how diverse items a user experiences has been importantly considered from a user perspective. Unfortunately, a large body of the research has been restricted to the static recommendation, which only investigates the items in a single recommendation.

More recently, there have been attempts to improve diversity in *session-based* or *sequential recommendation systems* (Wang et al., 2021) where a user sequentially selects items from the consecutive recommendation in a session. For dynamic recommendation, a sequential recommendation system exploits the history of the selected items to recommend next items. For example, the recommendation system adopted in YouTube chooses and displays the videos for recommendation based on the videos that a user has watched recently (Beutel et al., 2018). The recent studies on sequential recommendation systems have revealed that recently-consumed items in a session play more important roles in sequential recommendation (Jing & Smola, 2017; Tang & Wang, 2018).

In line with these studies, this paper proposes a deep learning model to estimate *diversification scores* of each item, which can indicates the degree of the potential contribution to users' diversity experiences. To this end, we first model all the potential items to be recommended and their relations by sequential recommendation as a *video recommendation network*, whose nodes and edges are videos and the recommendation relations among the videos, respectively. Considering the diversity from three perspectives – genre, publisher, and content semantics – we calculate the diversification score of a node from each perspective. The prediction model on the diversification scores adopts multiple graph

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neural network (GNN) layers with an extended attention mechanism that exploits the edge features among the nodes. Using the videos and their recommendation lists collected from YouTube, we evaluate the proposed model based on the synthesized user traces generated by the simulation experiments. Our evaluation demonstrates that the proposed model can estimate the diversification scores (of items) with high accuracy. By conducting an ablation study, we also reveal that the key features of the proposed model help improve the performance. Lastly, we implement an application to evaluate how users' experiences on diversity can be improved in a sequential recommendation system.

We highlight the main contributions as follows.

- We propose a deep learning model to estimate the diversification scores of individual items in video recommendation network, in terms of genre, publisher, and content semantics. To the best of our knowledge, this is the first study to predict degree of contribution of each item to a user's diversity experience. The proposed model adopts a message-passing mechanism of graph neural network with an extended attention mechanism to extract the information of the related videos for a given item in video recommendation network.
- We construct video recommendation network by collecting a large scale dataset from YouTube and conduct random-walk experiments to simulate user traces on the recommended items to evaluate the proposed model. We will make the collected dataset publicly available.
- To shed light on how to use estimated diversification scores, we implement an application that changes the rank positions of the items in the recommendation list in such a way that the items with the higher diversification scores are placed at the higher positions in the list. The evaluation, with the assumption that the items at the higher rank (i.e., higher position) have more opportunities to be selected, shows that the users' experience on the genre, publisher, and content diversity can be improved in a sequential recommendation system.

2. Related work

2.1. Diversification in recommendation systems

Supporting diversity has been considered as an important feature in online platforms from information retrieval (Abdool et al., 2020; Nie et al., 2022; Zhou et al., 2020) to recommendation systems (Adomavicius & Kwon, 2012; Antikacioglu & Ravi, 2017) as the lack of providing diversity can cause severe social problems such as popularity biases (Su et al., 2016), filter bubbles (Pariser, 2011) or echo chambers (Anderson et al., 2020). The growing importance of diversity in recommendation systems has attracted the research community to measure and improve the diversity from a service provider's and a user's perspectives. From the perspective of service providers, Adomavicius and Kwon (2012) suggested the *aggregate diversity*, which is the total number of items that have been recommended at least once over all the items, and proposed the ranking techniques and graph theoretic approaches to maximize the aggregate diversity. The popular metrics for the skewness of the distribution have also been used as a measure of diversity. For example, the Gini index is calculated in Ren et al. (2014), while the entropy is taken into account in Szlávik et al. (2011). Antikacioglu and Ravi (2017) proposed a recommendation algorithm to increase the sales diversity (which includes the aggregate diversity, Gini index, and the entropy) and rating quality based on a subgraph selection problem.

On the other hand, there have been another line of work to investigate diversity from the user perspective, which measures how diverse items individuals experience in the recommendation list (Nguyen et al., 2014; Zanitti et al., 2018; Zhang et al., 2012; Ziegler et al., 2005). In Ziegler et al. (2005), the Intra-List Similarity (ILS) is proposed, which measures the similarity among the items in the recommendation lists, by computing the pairwise similarity among the recommended books. Adopting the ILS as a target metric, Zhang et al. (2012) proposed the Auralist framework to improve user satisfaction in music recommendations. In Zanitti et al. (2018), the user-level diversity metrics in the above are adopted, and the category diversity of a user is maximized in movie recommendations. While in-depth studies on diversity have been carried out, they usually analyzed a static recommendation list, and relatively little attention has been paid to measure the diversity in sequential recommendation systems that dynamically adjusts items to be recommended as a target user keeps on consuming items in a session.

2.2. Session-based recommendation systems

Session or sequence-based recommendation systems aim to capture a user's dynamic preferences from her recent choices of items. Jing and Smola (2017) proposed an LSTM-based model that learns the recurring user activities for the survival analysis and the next item prediction in music recommendations. In Tang and Wang (2018), they proposed a Convolutional Sequence Model that transforms a user's recent actions into an image in terms of time and latent dimensions, and showed that its performance has been increased in the movie recommendations. Note that the approaches considering users' recent activities are adopted in the real-world platforms such as YouTube. Beutel et al. (2018) presented that YouTube makes use of a user's recent activities by using a recurrent neural network model with additional contextual data such as time delta between her activities.

While the recent studies on session-based recommendation systems have mostly focused on producing accurate recommendations based on the sequence of the selected items, relatively little work has been done to improve the diversity. In this work, we propose a deep learning model to predict the diversification score of an item, which can be used to improve a user's diversity experience in a sequential recommendation system.

3. Preliminaries

In this section, we describe three key concepts on which the proposed model is based: (i) a video recommendation network, (ii) genre, publisher, and content diversity, and (iii) diversification scores of an item in terms of genre, publisher, and content diversity.

Video recommendation network A video recommendation network models the relations among videos by recommendation as a directed graph $G = (V, E, W)$, where V, E are the sets of the nodes (i.e., videos) and edges, each of which connects from node i to node j when item j is in the recommended list of item i, respectively. The weight $w_{ij} \in W$ of the edge from i to j is a vector that represents edge features. In this paper, we build the video recommendation network by collecting the recommended videos in YouTube, which is described in Section 4.

Genre, Publisher, and Content Diversity The diversity of user experiences indicates *how diverse videos a user has watched in a session or a sequence*. To quantify the diversity of a user's experience, we first define diversity score as the average distances of all the pairs in a given sequence, whose formal equation is as follows:

$$Diversity = \frac{\sum_{\forall i,j \in V} \sum_{i \neq j} d(i,j)}{n * (n-1)} \quad (1)$$

where V and $d(i,j)$ are the set of videos that a user has watched, and the distance function between videos i and j, respectively. Note that this approach is similar to *Intra-List Similarity* (ILS) (Ziegler et al., 2005), a popular method to compute the similarity of items in a recommendation list. We only replace the similarity function in ILS with the distance function. The diversity of a given list is in a range of [0, 1] and reaches 0 when all the videos in the list belong to the same group, while exhibits 1 when the user has watched the videos that

belong to mutually different groups. Here a group can be a category, a publisher or a set of similar content properties.

We next define three types of diversity in terms of genre, publisher, and content perspectives by differentiating the distance function in Eq. (1). Following the definition in the prior studies (Dash et al., 2019; Vargas et al., 2014), we measure how diverse genres/categories of items are experienced by a user, denoted as *genre diversity*. We also provide a notion of *publisher diversity*, which compute how diverse publishers of videos are watched by a user. For distance functions for both diversity metrics, we use a binary function which outputs 1 when the genres and publishers/channels of the two videos are different, and 0 otherwise. The distance function for content diversity is computed based on the comparison of the TF-IDF vectors of two videos. To this end, we first use combine textual descriptions, title and the *tag*, the last of which is either a word or a phrase that attached to the video by an uploader to describe its content semantics. Considering the combined text and the tokens therein as a document and words for a video, we compute a TF-IDF vector of the given video. We then calculate the distance between two videos (i and j) as the cosine distance of the TF-IDF vectors of the videos, which is defined as follows:

$$d_{ij} = 1 - CosineSimilarity(TF-IDF_i, TF-IDF_j) \quad (2)$$

where $TF-IDF_i$ is the TF-IDF vector of video i. Note that the output of the distance function ranges from 0 (two videos deal with identical content) to 1 (two videos are entirely different in their semantics).

Diversification scores of videos The diversification score of a video is also differently defined from genre, publisher, and content perspectives. In particular, the diversification scores of node i for genre, publisher, and content diversity are defined as the average values of genre, publisher and content diversity of all the users who visit node i, respectively. The rationale of this approach is that a video in the watching history of a user may help to improve or degrade the user's diversity experience. Hence, assuming that the visit counts (or view counts) of all the nodes are sufficiently high, the nodes visited by more users with higher diversity experiences are more likely to help enhance the diversity experiences of the users who visit them.

4. Materials and methods

This paper seeks to develop a prediction model for the diversification scores of the videos in a sequential recommendation system. In this section, we describe our methodologies on (i) data collection and (ii) constructing a video recommendation network based on the collected dataset, and (iii) generating synthesized video watching sequences to compute genre, publisher, and content diversity. We then explain the proposed model with evaluation method in detail.

4.1. Data collection

To construct a video recommendation network, we first collect the videos with their recommendation relations from YouTube, the second most popular website[1] that provides diverse types of videos (e.g., movies, sports, or news) uploaded by individual users or organizations. To this end, we developed a data collection pipelining module leveraging the snowball sampling technique (Airoldi et al., 2016; Cheng et al., 2008; Dash et al., 2019). That is, the pipelining module first chooses a set of seed videos, and then collects the list of recommended videos of each video in the seed set. By iteratively crawling the lists of the recommended videos of the collected ones, the pipelining module can crawl all the videos associated by recommendations. In the below, we describe each stage of the data collection pipelining module.

[1] https://www.alexa.com/topsites

Collecting Seed Videos: At the first stage of the pipelining module, we find a set of seed videos by using the YouTube Data APIs,[2] through which we can collect the popular videos at the time of requests. Using the APIs, we first crawled the most popular videos of the 12 categories (e.g., Film/Animation, Music, Entertainment, etc.) for a week from Aug. 1, 2020 in the United States. Since the YouTube APIs provide no videos in the News category, we additionally collected the videos in News by capturing the videos in the *Trending News* page. Note that we repeated the seed video collection every hour during the crawling period and selected the top 10 videos in each category in terms of the number of appearance, which is a seed set of the given category.

Crawling Recommendation Lists: Starting with the seed set of the top 10 popular videos in each category, the pipelining module at the second stage collects the recommendation lists of each video for three times. Using selenium based crawlers, we fetch the recommendation list (of each seed video), which consists of 20 associated videos as shown in Fig. 1(a). Note that we have not included any user information (e.g., cookies) to avoid personalized recommendations, and furthermore, fetching the recommendation list is repeated for three times to mitigate the transient behaviors of recommendations.[3] We then aggregated all the newly collected videos and used them as a new seed set of the next iteration (shown in Fig. 1(b)). Here, we excluded the recommended videos in the different categories in the new seed set (for the next iteration) to simplify the crawling process. Since the number of videos in the seed set grows exponentially after each iteration, we iterate the crawling process only five times. Using the YouTube Data APIs for the collected videos, we additionally gathered each video's metadata including the title, description, and tags (a set of descriptive keywords attached to the video).

Table 1 summarizes the statistics of the dataset. In addition to the number of collected videos and channels (or publishers) in each seed category, we also calculate the average number of tags of each video and the portion of the recommended videos whose categories are the same as or different from the seed category. As shown in Table 1, at least 90K videos are collected in each category. Note that the portion of the recommendation videos in the same category is lower for Film & Animation and Comedy categories, indicating that the videos in the different categories are slightly more likely to be recommended for the ones in these categories.

Ethical Consideration on Data Collection: We acknowledged that our crawling process might affect the YouTube service performance. To minimize the overhead of the service, we restricted our crawlers to request only the text information of the videos and their recommendation lists and not to request any multimedia content (images or videos). For the other data resources such as initial seed videos or meta information, we used YouTube APIs with request rate limits.

4.2. Building a recommendation network

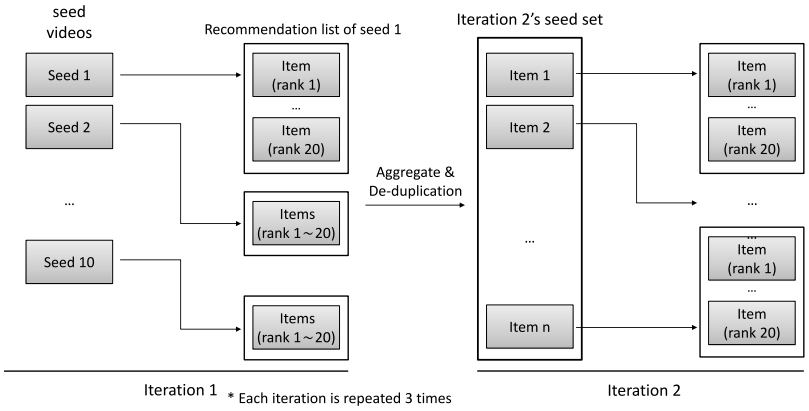
Based on the collection of video recommendations, we build the video recommendation network as defined in Section 3. Here, we use two values calculated from the collected dataset for w_{ij}: (i) the number of appearances of item j across multiple collections of the recommendation list of item i and (ii) the average of rank positions of item j across multiple collections of the recommendation list of item i. For instance, if a video j has appeared in video i's recommendation list for all three times during our crawling process, with each rank

[2] https://developers.google.com/youtube/v3/docs/videos/list

[3] To determine the appropriate number of repetitions, we performed a simple experiment. That is, for each of 1000 randomly chosen videos, we fetched the recommendation lists for 100 times and we found that the recommendations collected for three times included 80% of the unique videos collected from 100 times. Therefore, we decided to crawl the recommendation lists of each video for three times.



(a) Layout of YouTube player with the recommendation list



(b) Iterative crawling of recommendation lists

Fig. 1. YouTube video player layout and the data crawling process is illustrated.

Table 1
Basic statistics of the dataset is summarized.

Seed Category	Number of Collected Videos	Number of Channels	Avg. Number of Tags	Portion of Recommended Videos in the Same/Different Category
Film & Animation	172,550	49,608	17.45	48.8%/51.2%
Autos & Vehicles	483,920	119,283	18.53	75.1%/24.9%
Music	410,574	124,508	12.16	74.3%/25.7%
Pets & Animals	157,076	41,377	17.73	69.8%/30.2%
Sports	388,152	61,942	17.02	75.1%/24.9%
Gaming	239,292	25,748	18.88	71.2%/28.8%
People & Blog	263,011	63,211	17.40	53.8%/46.2%
Comedy	91,437	23,171	17.25	49.5%/50.5%
Entertainment	401,322	75,469	17.29	58.9%/41.1%
News & Politics	304,364	49,815	15.33	57.8%/42.2%
Howto & Style	227,102	46,379	13.89	65.3%/34.7%
Education	154,280	35,808	16.98	57.7%/42.3%
Science & Technology	268,719	59,575	17.16	64.7%/35.3%

as (3, 5, 4) (1st being the highest ranking, uppermost position), w_{ij}'s value would be [3, 4]. Throughout the described process, we finally obtained a video recommendation network consisting of 611,090 nodes and 12,690,030 edges.

4.3. Generating synthesized video watching sequences

Since it is difficult to gather the real-world user traces in the constructed video recommendation network (i.e., a sequence of videos that users consumed when they followed recommendations), we conduct a random walk-based simulation, which have been popularly used to mimic the history of the user visits on recommendation network (Dash et al., 2019; Papadamou et al., 2019; Roth et al., 2020). That is, starting at a point (say, node i), a user moves to one of its neighbors, meaning that the user selects and watches one from video i's recommendation list. By iterating this process until the termination condition is met, the random walk process finally generates the sequence of nodes simulating the user's watching session. The four components of the random walk process on the recommendation network are as follows:

- **Starting point**: The starting point indicates which video a user watches at first. We let a random walker start from every single node in the recommendation network with the rationale that it is hard to specify a user's first location since she can start from any video.
- **Transition**: A user's transition from a node to another depends on various factors such as the relevance or the rank position in the recommendation list. One of the key factors for the user choice is the rank positions of the items in the recommendation list, as

reported in Zhou et al. (2010). In line of this, we assume that a user selects a neighbor (video) in proportion to its rank position. That is, if the rank of a neighbor is the higher than another, it will be chosen with the higher probability. Hence, we define the transition probability from node i to node j as follows. $p_{ij} = w_{ij} / \sum_{w \in W_i} w$, where w_{ij} and W_i are the weight from node i to j and the set of weights of the outgoing edges from node i, respectively. We define weight w_{ij} as $1/log_2(rank_{e_{ij}} + 1)$ where $rank_{e_{ij}}$ is the averaged rank of e_{ij}, inspired by Discounted Cumulative Gain (DCG) (Järvelin & Kekäläinen, 2002) that measures the ranking quality of the recommendation. Note that during a random walk, we penalize the user re-visiting the same node by multiplying a small scaling factor ϵ into w. In this work, we hypothesize that a user will avoid consuming the same item again during a session.[4]

- **Teleportation**: While teleportation is a mechanism that a random walker jumps to another node that is not a neighbor, we assume that there is no teleportation since such teleportation should reflect the user's own interest or personalized choice model, which is out of scope.
- **Termination condition**: Like the prior studies (Dash et al., 2019; Röchert et al., 2020), we use the number of steps (i.e., number of transitions) as a termination condition. While prior studies used a fixed number of steps, the number of steps is taken uniformly between [5, 100] since there is no consensus in the number of steps.

Based on the components described above, we simulate 611,090 random walkers in an experiment; each walker imitates a user starting at each node in the recommendation network. In other word, we obtain 611K sequences of the videos in an experiment. Note that we performed total 20 experiments and aggregated all the sequences to measure the genre, publisher, and content diversity.

4.4. Problem statement

The goal of the proposed model is to estimate the diversification scores of node i in terms of genre, publisher and content diversity, denoted as $S_{i,gen}$, $S_{i,pub}$, and $S_{i,con}$, respectively. Note that we develop separate models to predict the diversification scores for genre, publisher, and content diversity.

4.5. Model architecture

Fig. 2 shows the overall architecture of the proposed model that predicts a node's diversification score in the video recommendation network, which consists of three layers: (i) node feature embedding, which computes the fixed-size vector representation of the embedding of a given node, (ii) embedding propagation, which passes a node embedding to its neighbors and updates the embeddings of the associated nodes and edges, and (iii) prediction, which estimates the diversification scores of the nodes using the updated node embeddings.

4.5.1. Node feature embedding

The node feature embedding layer computes the representation vectors from the node features that may affect users' diversity experiences. In particular, the embedding layer takes a feature vector of a node and outputs an embedding vector, calculated by multiplying a trainable weight matrix with the feature vector. Formally, the process of generating an embedding vector of node i is defined as follows.

$$h_i^0 = W_{embed}\, r_i \tag{3}$$

where r_i is a feature vector of node i, W_{embed} is a trainable weight matrix and h_i^0 is the computed embedding of node i. Note that we use one-hot encoding of a video's category and channel as the feature vectors of the prediction model on diversification scores for genre and publisher diversity, respectively, while a TF-IDF method is used for the feature vectors for content diversity.

4.5.2. Embedding propagation

The relations by recommendations, which are the edges in the video recommendation network, may be linked to users' diversity experiences since the users select a video from the recommended items in a session (or sequence). For example, if most of the neighbors of a video (either recommending or being recommended) have the same genre, we can expect that the diversification score of the given video (or node) for genre diversity would be low. Considering this, the proposed model uses not only the node features, but also the recommendation relations. To this end, we adopt the embedding propagation layer which is based on the message-passing mechanism employed in the popular graph neural network (GNN) models like Graph Convolution Network (GCN) (Kipf & Welling, 2017) and GraphSAGE (Hamilton et al., 2017). That is, the embedding propagation layer accepts and updates the embedding of every node, which is propagated to its neighbors.[5] The layer then updates the embedding of a node by aggregating the information (i.e., the embeddings of its neighbors) propagated from its neighbors along with the information of the node itself. Note that we stack multiple embedding propagation layers to consider multi-hop neighbors (e.g., neighbors of the neighbors) to update node embeddings.

The edge weights of the relations among the nodes in video recommendation network are heterogeneous, implying that different neighbors can affect a diversification score of a given node in a different way. To consider this, we propose *feature-augmented attention* mechanism, which augments the feature vectors of the neighboring edges to compute attention scores. In particular, the proposed model first initializes a 4-dimension embedding vector of an edge based on (i) the number of appearances in recommendation list, (ii) the average rank positions in recommendation, (iii) distance ($d(i,j)$), and (iv) the closeness ($1 - d(i,j)$) between two nodes. The proposed model then concatenates node embeddings and edge embeddings to compute the attention scores of the individual neighbors. Note that the edge embeddings are iteratively updated based on the associated node embeddings during the embedding propagation process.

The process of the embedding propagation is defined as follows. Here, we omit the bias term in the equations for brevity.

$$h_j^{l+1} = \sigma\Big(\sum_{i \in \mathcal{N}(j)} \alpha_{j,i,e} W_N^l h_i^{(l)}\Big) \tag{4}$$

where h_j^l is node j's node embedding (also called hidden representation) at layer l, $\mathcal{N}(j)$ is the set of neighbors of node j, W_N^l is the trainable weight matrix for the node embedding to apply the shared linear transformation. The attention coefficient $\alpha_{j,i,e}$ is calculated as follows:

$$\begin{aligned} \alpha_{j,i,e} &= \mathrm{softmax_j}(x_{j,i,e}) \\ x_{j,i,e} &= \mathrm{LeakyReLU}(\vec{a}^T [W_N h_j^l \| W_N h_i^l \| W_E h_e^l]) \end{aligned} \tag{5}$$

where h_e is the embedding of edge e from node i to node j, W_E is the trainable weight matrix for the edge embedding, and $\vec{a}$ is the trainable weight vector. The embedding propagation layer updates the edge embedding h_e as follows:

$$h_e^{l+1} = W_E h_e^l \tag{6}$$

where W_E is the same weight matrix used to compute attention coefficients.

[4] We suppose repetitive consumption of the same item rarely happens during a session to simplify the simulation.

[5] While the recommendation network is directed, the embeddings are propagated regardless of edge directions.

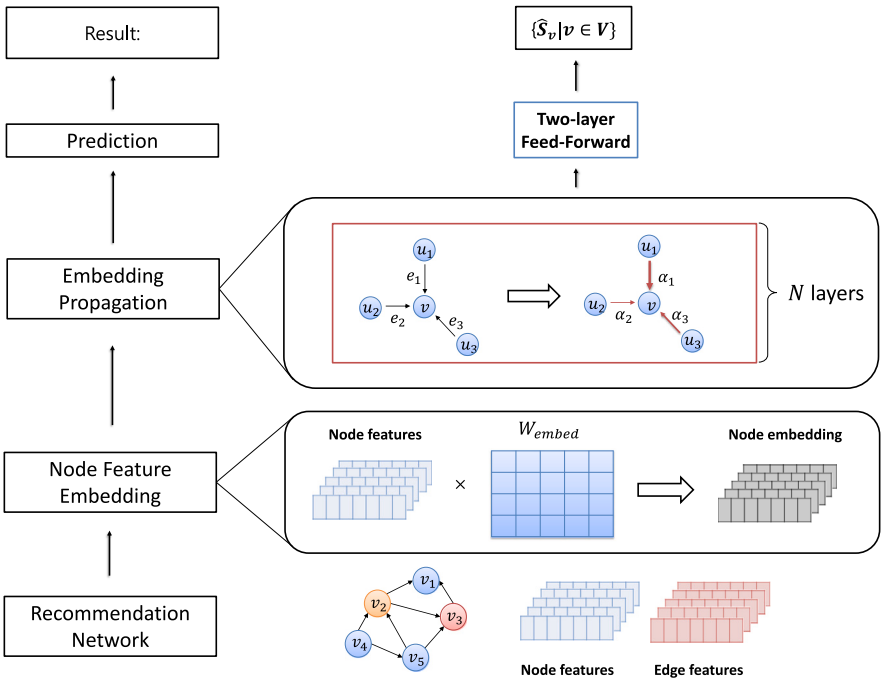


Fig. 2. An illustration of the model and its components is shown.

4.5.3. Prediction

Using the representations of all the nodes $(h_1, h_2 \ldots, h_k)$ from the embedding propagation layers, we predict the diversification scores of a node at the prediction layer that consists of two fully-connected layers. Here k is the number of nodes in the recommendation network. Formally, the process is expressed as follows:

$$\hat{S}_i = W_2(\sigma(W_1 h_i)) \tag{7}$$

Note that we use the Mean Absolute Error (MAE) that computes the error between the predicted score and the observed score as a loss function, which is defined as follows:

$$L_{MAE} = \frac{1}{k}\sum(\hat{S}_i - S_i) \tag{8}$$

5. Experiments

5.1. Evaluation methodology

We selected 397,811 nodes out of 611,090 nodes, which are visited by more than 50 random-walkers, as the target nodes. Since the diversification score of a node is calculated as the average of the diversity scores of visiting users, only the node visited by a sufficient number (i.e., 50) of random walkers is used to obtain the stable value. The set of the target nodes was then split into training, validation, and test sets with 6:2:2 ratio. We use the Mean Squared Error (MSE) and R^2 as the evaluation metrics. Note that we conducted the experiment five times with the different data split, whose averages are reported.

5.2. Baseline models

For a comparison purpose, we also measure the performance of several baseline models, which can be grouped as feature-based and GNN-based models. The feature-based models use hand-crafted features characterized by considering the local structure of a given video in recommendation network. In particular, we compute the six common features and separate feature vectors to predict diversification scores for genre, publisher, and content diversity, as described in Table 2.

Table 2
The hand-crafted features used in the baseline model are described.

Feature Type	Description
Common	Unweighted degrees
	Weighted degrees in terms of number of appearances described in Section 4.2
	Average rank of a given node in the recommendation list for its neighbors
	Average rank of the neighbors in the recommendation list for a given node
	Degree centrality
	Eigenvector centrality
Genre Diversity	The concatenated vectors of the category distribution of in-/out-neighbors at 1, 2, and 3 hop counts
	One-hot vector of the category of a given node
Publisher Diversity	The concatenated vectors of the publisher distribution of in-/out-neighbors at 1, 2, and 3 hop counts
Content Diversity	The number of in-/out-neighbors whose cosine distances of TF-IDF vectors to a given node in the first, second, third, and fourth quartiles of all the distribution of the cosine distances in recommendation network

We also implemented 4 GNN-based deep learning models: (i) GCN (Kipf & Welling, 2017), (ii) GraphSAGE using the mean and max-pooling as aggregate functions (Hamilton et al., 2017) (denoted as SAGE(MEAN) and SAGE(MAX), respectively), and GAT (Velickovic et al., 2018).

5.3. Implementation environment and hyperparameters

All the models including the proposed model and other GNN-based baseline models are implemented with Deep Graph Library (v0.7) with PyTorch (v1.7.1) backend on the machine of Intel Xeon 2.1 GHz CPU with NVIDIA Tesla V100 GPU. We conducted grid-searching to select the hyperparameters of the proposed model, and selected the ones showing the best performance on the validation data. After the observation, we finally set the number of layers and the dimension

Table 3
The experiment results of the baseline and proposed models are shown.

Types	Models	MSE ↓ ($\times 10^{-4}$)			R^2 ↑		
		S_{gen}	S_{pub}	S_{con}	S_{gen}	S_{pub}	S_{con}
Baseline	SVR	453	445	398	0.8138	0.5897	−0.6711
	GCN	21.2	3.25	1.99	0.8870	0.9640	0.9504
	SAGE (MEAN)	19.0	4.40	2.23	0.8990	0.9513	0.9434
	SAGE (MAX)	164	13.0	5.26	0.1265	0.8561	0.8688
	GAT	15.3	3.74	2.07	0.9210	0.9587	0.9488
Proposed	excl. NE	18.7	**3.01**	5.84	0.9004	**0.9666**	0.8399
	excl. NE	14.9	3.70	1.96	0.9203	0.9591	0.9512
	All	**14.0**	3.27	**1.95**	**0.9251**	0.9638	**0.9513**

of the edge embedding for embedding propagation as 3 and 16, respectively, while the dimensions of output features for node feature embedding and each embedding propagation layer are set to 16 and 64, respectively. In the prediction layer, the dimensions of output features of two fully-connected layers are set to 32 and 1 with ReLU as an activation function. For training, we use Adam optimizer and set the learning rate, dropout probability, and epoch count to 0.001, 0.1, and 10,000, respectively. In addition, the early-stopping mechanism, which finishes training before reaching the epoch count when the performance improvement is marginal, is adopted with 500 patience.

6. Results

6.1. Overall performance

Table 3 shows the experiment results. Overall, the proposed model outperforms all the baseline models for the genre diversity (S_{gen}); the R^2 value of the proposed model for S_{gen} is 0.9251 while those of SVR, GCN, SAGE (MEAN, MAX), and GAT are 0.8138, 0.8870, 0.8990, 0.1265, and 0.9210, respectively. When we compare the performance of the proposed model with GCN, SAGE (MEAN), and GAT, the MSE value has been improved by 50%, 35%, and 7%, respectively. These results show that the proposed model can accurately estimate the nodes' diversification scores for genre diversity. Performing the T-test on squared errors for the GAT and the proposed model, we confirm that the result is statistically significant (p-value < 0.01). Note that the proposed model shows a slightly lower performance than GCN in terms of S_{pub}, with R^2 value of the proposed model is lower by 0.0002. The p-value of the same T-test between these models is 0.5, which confirms that performance of two models is not significantly different.

Furthermore, the performance of GNN-based models is much higher than SVR model, indicating that GNN-models can extract comprehensive features from the relations in recommendation network than the models based on hand-crafted features. Among GNN-based models, GAT shows the best performance in terms of S_{gen} while the MSE of GCN is the lowest in terms of S_{pub} and S_{con}. As each baseline model focuses on the different aspects for the given graph,[6] these results imply that employing the attention mechanism to reflect the relations among neighbors is a key factor to estimate nodes' diversification scores for S_{gen}. Similarly, we conclude that the features of the neighbors are more important in predicting S_{pub} and S_{con} than the attention mechanism.

6.2. Ablation study

We further investigate how each design choice of the proposed model has an effect on the prediction of nodes' diversification scores. To this end, we conduct the ablation study on the node embedding and on updating edge embedding. We also observe the impact of the size of the edge embedding.

Table 4
The impact of edge embedding size ($|h_e|$) in predicting S_{gen} is shown.

Edge embedding size	MSE ↓	R^2 ↑
k = 4	1.43×10^{-3}	0.9238
k = 8	1.40×10^{-3}	0.9252
k = 16	1.37×10^{-3}	0.9269
k = 32	1.42×10^{-3}	0.9240

The impact of node feature embedding We first investigate the impact of embedding the node features. To this end, we measure the performance of the proposed model by excluding the node feature embedding layer, and instead use random vectors as the node features, which is labeled as "excl. NE" in Table 3. Clearly, the performance of the model without node feature embedding is worse than the original proposed model for S_{gen} and S_{con}, meaning that the embedding layers of node features help to predict nodes' diversification scores. On the other hand, for S_{pub}, the model without the node embedding layer outperforms the original proposed model and other baseline models including the model with GCN layers, which implies that the node feature embedding layer degrades the performance. We also observed the same pattern in the case of GCN; the MSE of the GCN without the node feature embedding layer for S_{pub} is 3.21×10^{-4}, which is lower than that of the GCN with node feature embedding (3.25×10^{-4}). This may be due to the fact that nearly 60% of the channels (56k out of 90k) have only a single video. Thus, the node embedding layer fails to learn channel-level information.

The impact of updating edge embedding The proposed model updates the edge embedding at each layer by multiplying the edge embedding with W_E. We next explore whether updating the edge embedding across the layers helps to estimate the nodes' diversification scores by measuring the prediction performance of the proposed model without the updated edge embedding. To this end, we modify the proposed model not to propagate the updated edge embedding across the layers; instead, the proposed model uses raw-level edge features at each graph layer. The performance of the model excluding edge embedding is shown as "excl. EE" in Table 3, which indicates that the modified model without updating edge features achieves the lower performance for every diversity type. Note that even the modified model still shows better performance than GAT, indicating that using the raw-level edge features can improve the performance.

The impact of edge embedding size We investigate how the size of the edge embedding between embedding propagation layers affects the model's performance by measuring the MSE and R^2 by changing the size of the edge embedding from $|h_e| = 4$ to 32. Here, we show the result on predicting a node's diversification scores only for the genre diversity. As shown in Table 4, R^2 increases as $|h_e|$ increases up to 16, and then decreases. This result shows that using the larger size for the edge embedding can improve the performance to a certain degree, but increasing the size excessively may introduce overfitting and degrade the model's performance.

7. Potential application

In this section, we further explore how the proposed model can be used to enhance users' diversity experiences. In particular, we investigate how the diversity score estimated by the proposed model can cooperate with a re-ranking method, in which a recommendation system changes the ranks of the recommended items for display, so that the items with higher diversification score to be located at the top of the recommended list. Note that re-ranking is a popular way to measure and improve diversity in recommendation systems (Antikacioglu & Ravi, 2017; Belém et al., 2016).

[6] GAT adopts an attention mechanism to differently weigh on the relations between a given node and its neighbor, while GCN and GraphSAGE assume all the neighbors have the similar importance

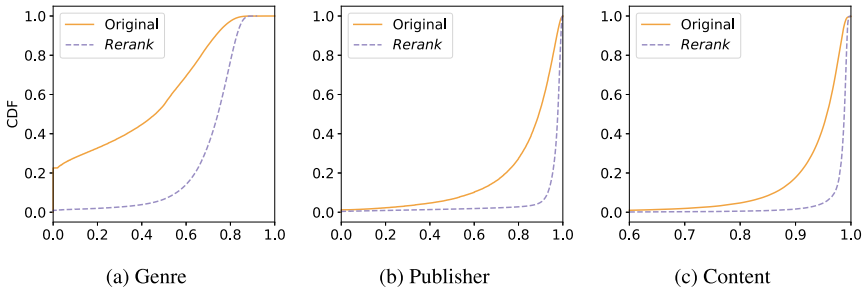


Fig. 3. CDF of random walkers' diversity of original/re-ranked simulation is plotted.

To employ the diversity score by the proposed model into a re-ranking method, we first compute a node's diversification score for each diversity type with the best model among the proposed one and the baseline ones. Then, we re-rank the neighbors of each node by the predicted diversification scores in the descending order so that a neighbor with the higher score is placed higher in the recommendation list. Using this re-ranking, we conduct the same random-walk simulation and compare the distribution of the diversity scores of walkers. Fig. 3 plots the distributions of walkers' genre, publisher and content diversity from the simulations of the original and the re-ranked recommendation lists. We observe that walkers' diversity experiences on the re-ranked recommendation network are improved across all the diversity types. These results reveal the potential of predicting the diversification scores, which can help enhance the users' diversity experiences. Note that the simple re-ranking technique is only used to demonstrate its potential and far from optimal.

8. Concluding remarks

We proposed a deep-learning model to predict nodes' diversification scores, which represents how much an item can affect users' diversity experiences. Building the video recommendation network from the collected dataset from YouTube, we carry out the random walker-based experiments to compute a node's diversification score. Our evaluation on the proposed model demonstrates that diversification scores for genre, publisher, and content diversity can be used to enhance the diversity experiences. Furthermore, the ablation study shows that the design choices of the proposed model (e.g., node feature embedding) contributes to improving the prediction performance. By employing diversity scores estimated by the proposed model into a re-ranking method, we demonstrate that the diversity scores can improve users' diversity experiences, which can provide a valuable insight to improve recommendation systems in practice.

Limitations and Future work: Despite the significant contributions of this paper, this work is restricted by a few limitations. First, the random walk simulation to generate user traces in recommendation network is designed to mimic general user behaviors, which can be different from user behavior in practice. Thus, designing more comprehensive algorithms for random walk simulation (e.g., adopting user preference to transition probability) or using the user traces in real-world can provide more valuable insights on providing diversity in recommendation network, which will be conducted in future work. Second, the evaluation in this work relies on the collected datasets from YouTube. Thus, we caution against generalizing the methodology and findings to other platforms or other datasets.

CRediT authorship contribution statement

Selin Chun: Conceptualization, Methodology, Data curation, Investigation, Writing – original draft. **Jinyoung Han:** Conceptualization, Writing – review & editing. **Daejin Choi:** Supervision, Writing – review & editing. **Taekyoung Kwon:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset and all the implemented source codes that support the findings of this study are available at https://sites.google.com/view/aidml/code-datasets/eswa23.

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